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Project Title:	Dynamics of auxin transport protein localization and gravitropism		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
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Space Biology Special Category:	None		
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Contact Monitor:		**Contact Phone:**	
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Task Description:

Plants respond to gravity in numerous ways; the best understood is the change in growth orientation in response to reorientation of plants relative to the gravity vector. A growth asymmetry develops, which allows the plant organ to return to the original angle of growth relative to gravity. Understanding gravitropism is essential to facilitate growth of plants in the microgravity environment of space. Gravitropic bending is mediated, at least in part, by redistribution of the plant hormone auxin. This proposal examines the mechanisms by which the cellular asymmetry of auxin transport proteins is established and the molecular mechanisms that allow translocation of the auxin signals from the site of perception to the site of reaction.
We will employ three approaches to transiently alter the expression and localization of auxin transport proteins to examine the effect on gravitropic curvature in roots of the model species Arabidopsis thaliana. First, we will use a temperature sensitive mutant, scd1-1, with a defect in endomembrane targeting, to transiently disrupt, auxin transport protein localization and gravitropic curvature.

Second, we will transiently perturb the activity and/or localization of auxin transport proteins through a pharmacological approach and measure the resulting effect on gravitropic kinetics. In our third approach, we will use well characterized null mutants with defects in auxin transport proteins and will express wild type copies of defective genes in an inducible expression system. This approach of reintroducing a gene function at a controlled time points during the gravitropic process, should allow us to understand when these proteins function in controlling auxin transport and gravitropism. The planned experiments will combine high resolution laser scanning confocal microscopy and high temporal resolution gravitropic analyses with transient perturbation of auxin transport to understand the dynamics of auxin transport protein localization and resulting auxin asymmetries that drive differential gravitropic growth.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

We have largely completed our examination of the scd1-1 mutant, which has a defect in a gene encoding a protein predicted to function in RAB GTPase-dependent protein trafficking, specifically a RAB-GEF (guanine exchange factor, which is needed to activate RAB protein activity). We found that IAA transport, the distribution of IAA efflux proteins, and gravity response were altered in the scd1-1 mutant at at the nonpermissive temperature of 25°C (Isley, 2011). In contrast, at the permissive temperature of 18ºC, these processes were at wild-type levels. The gravitropic response of scd1-1 seedlings was quantified by high resolution imaging, taking images every 2 minutes for 8 hours to measure the full gravitropic response. After a 90° reorientation relative to the gravity vector, the wild type plants grown at 25°C showed a complete gravitropic reorientation with a 90° root angle within 8 hours, while scd1-1 has a delayed gravitropic response reaching only a 45° root angle after 8 hours. We monitored gravitropic curvature and growth rate in scd1-1 as seedlings were switched from growth at 18°C to 25°C. The reductions in root gravitropic curvature in scd1-1 were evident in seedlings that were exposed to the nonpermissive temperature (25°C) for as little as 8 hours.
Second, we examined the transport of auxin in the scd1-1 mutant, to determine if the mechanism by which gravity response is impaired is tied to the ability to transport IAA. Auxin moves in two directions in the root in two distinct tissues. Basipetal or shootward IAA transport, regulates differential gravitropic growth, and acropetal or rootward transport, provides auxin for lateral root development. We examined both polarities of IAA transport and find that basipetal and acropetal transport are reduced in the scd1-1 mutant relative to wild-type at 25°C, while similar amounts of IAA transport were detected at 18°C. Just 6 hours of incubation at RT (25°C) after growth at 18°C was sufficient to show a difference in IAA transport. We found that the time required to alter basipetal auxin transport was similar to the time required to altered gravity response suggesting a physiological relationship between these two parameters.

The third line of experiments in scd1-1 focused on examination of the effects of temperature changes on the distribution of auxin transport proteins. We used LSCM to examine the tissue level and sub-cellular distribution of auxin efflux carriers ABCB4, PIN1, PIN2, and PIN3 that are translational fusions to GFP and the influx carrier AUX1 fused to YFP. When scd1-1 was grown at 25º C, the expression of the PIN1, 2, and 3 reporters was restricted to fewer cells and was reduced in intensity, and PIN2 and 3 accumulated in endomembranes. The subcellular distribution of these three reporters was similar in wild-type and scd1-1 at 18ºC with the expected asymmetrical pattern at the plasma membrane in wild type and scd1-1. AUX1 and ABCB4 fusions were unaffected in the mutant. Current experiments focus on real time imaging to watch the movement of PIN-GFP fusions from the membrane into endomembrane organelles in real time to understand the dynamics of this process.

The last set of experiments that is underway with scd1-1 is to identify the subcellular compartments that accumulate PIN proteins at the nonpermissive temperature. We have completed a colocalization analysis with FM464 and found that there is no colocalization of PIN2-GFP with FM464 stained endosomes (Isley, 2011). Therefore we are confident that SCD1 functions in targeting PIN2 to organelles other than the endosome. We have crossed endomembrane specific fluorescent markers for Golgi, endoplasmic reticulum, and vacuole into scd1-1x PIN1 GFP and scd1-1xPIN2GFP. Imaging is just beginning for these combination lines. If an endomembrane specific marker line shows fluorescence that colocalizes with the observed PIN: GFP endomembranes, this will provide insight into the role of SCD1 protein in targeting auxin transport proteins, which are then needed for gravitropic responses.

A second line of experimentation utilized a pharmacological approach to transiently disrupt activity and/or targeting of auxin transport proteins to understand their role in gravitropic responses. We examined the effect of four chemicals that disrupted gravity responses in a chemical genomic screen by Glenn Hicks and Natasha Raikhel at UC Riverside. We measured the effects of four of these compounds on gravitropism, root elongation, auxin transport, and asymmetric expression of DR5-GFP (an auxin responsive promoter driving GFP expression). These compounds reduced gravitropic curvature, but with a distinct set actions. Some alter auxin transport, some alter auxin signaling, and some affect other aspects of gravitropic signaling (Hughes, 2011).

The final avenue of experimentation is to construct inducible constructs expressing auxin transport proteins in mutants that do not synthesize these proteins. Tanya Falbel is leading this process with assistance from two undergraduates, Lorena Rios Acosta (University of Puerto Rico - Mayaguez) and Catherine Ballenger (University of Wisconsin - Madison). We will then monitor gravitropic curvature under different induction regimes, as well as monitor the levels of the auxin transport proteins at the times at which they are effective.

References

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Bibliography Type:	Description: (Last Updated: 08/25/2020)
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Dissertations and Theses	Hughes R. "Investigating root gravitropism in Arabidopsis thaliana and its inhibition using three endosidin drugs." Undergraduate Honors Thesis, Wake Forest University, May 2011. , May-2011